

An Adaptive Driver Support System: User Experiences and Driving Performance in a Simulator

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Objective: The aim of this study was to test the implementation of an adaptive driver support system.

Background: Providing support might not always be desirable from a safety perspective, as support may lead to problems related to a human operator being out of the loop. In contrast, adaptive support systems are designed to keep the operator in the loop as much as possible by providing support only when necessary.

Method: A total of 31 experienced drivers were exposed to three modes of lane-keeping support: nonadaptive, adaptive, and no support. Support involved continuously updated lateral position feedback shown on a head-up display. When adaptive, support was triggered by performance-based indications of effort investment. Narrowing lane width and increasing density of oncoming traffic served to increase steering demand, and speed was fixed in all conditions to prevent any compensatory speed reactions.

Results: Participants preferred the adaptive support mode mainly as a warning signal and tended to ignore nonadaptive feedback. Furthermore, driving behavior was improved by adaptive support in that participants drove more centrally, displayed less lateral variation and drove less outside the lane's delineation when support was in the adaptive mode compared with both the no-support mode and the nonadaptive support mode.

Conclusion: A human operator is likely to use machine-triggered adaptations as an indication that thresholds have been passed, regardless of the support that is initiated. Therefore supporting only the sensory processing stage of the human information processing system with adaptive automation may not be feasible.

Application: These conclusions are relevant for designing adaptive driver support systems.

Keywords: adaptive automation, head-up display, lateral control, mental workload, driving behavior

INTRODUCTION

Enabled by technology, fully automated cars driving around in real traffic are no longer science fiction. In 2010, Google announced that it is testing self-driven cars using trained safety drivers inside the cars to take over control when necessary (Lee, 2010; Thrun, 2010). At the time of the announcement, seven test cars had driven a total of 140,000 miles with occasional human intervention and 1,000 miles without any human involvement (Markoff, 2010). Only one car was involved in an accident, when it was rear-ended while waiting at a traffic light. In the same announcement, it was pointed out that an automated car allows its passengers to spend their time more efficiently, improves safety, and increases road capacity, since self-driving cars respond faster than humans, have a 360° perception, do not get distracted or fatigued, and so on.

Although sitting back and relaxing while the automated system handles vehicle control is appealing, it represents a fundamental change of the driver's role from an active driver to a passive supervisor, potentially resulting in less safety (Brookhuis & de Waard, 2007). This change is similar to what has been seen in other high-tech task environments. Although full automation has many advantages, including reduction of small errors and better handling of routine operations (Wiener & Curry, 1980), there can be serious risks to performance when a manual takeover is required, as a result of reduced situation awareness, manual skill erosion, and late detection of automation failures (Bainbridge, 1983; de Waard, Van der Hulst, Hoedemaeker, & Brookhuis, 1999; Endsley, 1995). The aforementioned example shows that even the impressive driving performance shown by the Google test cars does not allow a driver at present to sit back and fully engage in other activities.

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To prevent “out-of-the loop” problems, it has been proposed to automate (sub)tasks only when support is needed, known as adaptive automation (Byrne & Parasuraman, 1996; Miller & Parasuraman, 2007; Mulder, Dijksterhuis, Stuiver, & de Waard, 2009; Parasuraman & Riley, 1997; Scerbo, 1996, 2001). In contrast to those in nonadaptive automation, the tasks performed by an adaptive system, or the level of automation (LOA), are not fixed but may change in real time. Changing LOA could be handled by the human operator in an *explicit* task allocation loop, which would add the task LOA management to the workload of the human operator. In contrast, LOA management could also be handled by an automated decision module in an *implicit* loop (e.g., Tattersall & Fairclough, 2003). Ultimately, a system is envisioned that supports and interacts with the human operator in a way that reflects social relationships. For instance, it could resemble a team member or subordinate (e.g., Mulder, de Waard, Hoogenboom, Quispel, & Stuiver, 2008). In this kind of human-machine interaction, the machine has the authority to initiate changes in the number of automated subtasks, the LOA per subtask, and the type of support (e.g., Scerbo, 2001). This capability requires real-time assessment by the system to decide when support is needed.

When designing an adaptive system, one needs to establish what defines the necessity for adapting task automation. As such, several broad strategies have been suggested (e.g., Morrison & Gluckman, 1994; Parasuraman, Bahri, Deaton, Morrison, & Barnes, 1992). For example, aid could be triggered by critical events in the environment that are not directly linked to user behavior. In contrast, the operator’s psychophysiological signals could also be used to trigger task support. Another user-adaptive approach is to monitor task performance and change LOA to prevent unacceptable performance degradation (e.g., Rouse, 1988). It has also been suggested that the LOA be coupled to mental workload (MWL), here defined as “the reaction to demand; the proportion of the capacity that is allocated for task performance” (de Waard, 1996), which would ideally create a system in which the operator’s mental workload remains relatively stable (e.g., Parasuraman et al., 1992).

Stabilizing mental workload, by preventing long periods of both overload and underload, has great advantages for what we think should be the ultimate purpose of an adaptive system in safety-critical tasks: to protect the operator and other humans in contact with the system. According to the operator functional state framework, for example, operating with high MWL, even if performance levels are still acceptable at a particular moment, might warrant a change of automation since acceptable performance is increasingly difficult or even impossible to sustain in demanding working conditions (e.g., Hockey, 1997, 2003). These considerations imply that MWL assessment could be used to decrease the probability of passing thresholds for unacceptable performance levels.

Although MWL assessment could be used as a trigger for task support, some hysteresis should be present in the system (Hoogeboom & Mulder, 2004) in that not all MWL changes should result in a modified support level. Dead bands will likely be necessary to prevent excessive, rapid switching between levels of support. Also, the effectiveness of an adaptive system will partly depend on the operator’s long-term acceptance of the adaptive system. Therefore, operator preferences in timing might temporarily overrule triggers. Last, behavioral adaptations when operating an adaptive support system may become an issue. Human operators themselves control how much effort they will invest and may find new ways to perform their task that were not anticipated by the system’s designers, which may potentially result in reductions or even reversals of potential safety gains (e.g., de Waard et al., 1999; Lee, 2008; Parasuraman & Riley, 1997). For instance, operators might try to maximize support levels, even if there is no need to reduce effort investment (Hoogeboom & Mulder, 2004).

Although setting the adaptive properties of an automated system is a major design requirement, another challenge is deciding how to support the human operator. Although available support types depend greatly on the task, some general concepts are available in the literature. The simple binary approach of manual versus fully automated systems was followed by a graded classification of LOA, including levels

such as the computer “allows the human a restricted time to veto before automatic execution” and “the computer offers a complete set of decision/action alternatives” (Parasuraman, Sheridan, & Wickens, 2000, p. 287). Moreover, LOA may also be viewed as a unidimensional continuum of automation degrees (e.g., Parasuraman, Sheridan, & Wickens, 2000).

A conceptual shift away from a decision and action execution-oriented approach to LOA was to include the early stages of the human information processing system, sensory processing and perception or working memory, as candidates for automation (Parasuraman et al., 2000). Another scale of automation degrees was presented in Flemisch, Kelsch, Löper, Schieben, and Schindler (2008). On this scale, the term *assistance* is used when most of the task is performed by the human operator and could include support types that are not readily included in the concept of automation, such as providing information, warnings, and advice. Conversely, at the other end of the scale, the term *automated* is used when most of the task is automated.

One example of a task for which support could be provided is steering a vehicle. Lane keeping (steering or tracking) is a major, but relative easy, part of maintaining safe control of a vehicle (Parkes, 1991). However, the relative ease with which drivers generally keep in their lanes could be contrasted with accident statistics. For instance, the United Nations Economic Commission for Europe (2007) report on statistics of road traffic accidents in Europe and North America shows that about one third of the accidents involving personal injury or death may be related to inadequate lateral control.

Steering is usually carried out in a highly automated fashion, although a driver will direct attentional resources to the task whenever the situation demands it and, in doing so, switch from the control level of operation to the maneuvering level (Michon, 1985). In demanding conditions, such as decreased visibility, narrow lanes, or when drivers are fatigued, it may be difficult to estimate lateral position (LP) adequately and therefore to maintain safety or comfort zones (Summala, 2005; Summala, Nieminen, & Punto, 1996). In these instances, a driver may benefit from more accurate and reliable LP information. Providing accurate and

reliable information would be an example of assisting the input stages of the human processing system (Parasuraman et al., 2000) and would constitute a relatively low degree of automation (Flemisch et al., 2008).

However, using raw (not averaged) data to trigger support, such as done by traditional lane-departure warning systems, may make a support system too reactive. Each driver displays naturally occurring and unsystematic driving behavior variations, which may be a primary reason for drivers to maintain safety margins (Brehmer, 1990; Ranney, 1994). A single (small) driving error does not necessarily imply that the driver needs support immediately, but frequently occurring minor errors is a stronger indication that driver support is needed. Hence, time intervals may be used to create “trigger variables.” In the case of the steering task, frequent or long periods of driving near the driving lane delineation could be an indication that not enough effort is being invested in keeping the car in the center of the lane. Another indication of inadequate effort investment would be lane departures, since these actions may set a vehicle in the direct path of oncoming traffic or on a potentially dangerous shoulder.

Last, the standard deviation of the LP (SDLP) can be used to assess changes in MWL. For example, the SDLP turned out to be a sensitive measure for overall driving capacity in experiments involving drugs and driving (O’Hanlon, Haak, Blaauw, & Riemersma, 1982; Ramaekers, Robbe, & O’Hanlon, 2000). Also, sleep deprivation and fatigue as a result of prolonged driving has been shown to increase SDLP (Anund, Kecklunda, Vadeby, Hjälm Dahl, & Åkerstedt, 2008; de Waard & Brookhuis, 1991), and in a previous study, SDLP was sensitive to changes in lane width and oncoming traffic (Dijksterhuis, Brookhuis, & de Waard, 2011).

To investigate the behavioral effects of providing support related to lateral control and to assess user experiences, we conducted a study in which a support system was introduced to provide objective information related to the vehicle’s LP, similar to a speedometer, by means of a head-up display (HUD). A HUD was chosen to keep the gaze directed toward the road as much as possible (e.g., Tufano, 1997). The HUD provided the driver with a continuous



Figure 1. Lane position information projected on the windshield of the simulator car.

flow of information on lane width and LP (see Figure 1 for a screenshot of the HUD). To investigate support effects across a range of lateral control demands, lane width and oncoming traffic density were varied. In addition, an adaptive driver support condition was created by (de)activating the HUD automatically when exceeding lateral control performance thresholds.

METHOD

Participants

We recruited 32 participants through poster announcements and paid them €20 for participating. However, 1 participant was excluded because of insufficient driving experience, leaving 31 participants (26 male). Ages ranged from 23 to 44 years ($M = 26.1$, $SD = 4.4$), and participants had held their full driving license for 5 to 24 years ($M = 7.5$, $SD = 3.6$). The self-reported total mileage ranged from 12,500 to 1,200,000 km (median = 40,000), and the annual mileage ranged from 2,500 to 50,000 km (median = 6,000). Finally, none of the participants reported using prescribed drugs that might affect driving behavior.

Simulator and Driving Environment

The study was conducted using an ST Software driving simulator consisting of a fixed-base vehicle mock-up with functional steering wheel, indicators, and pedals. The simulator was surrounded by three 32-in. diagonal screens, with each screen providing a 70° view of the driving environment. A detailed description of the driving simulator used can be found in Van Winsum and Van Wolffelaar (1993).

For the experiment, a route was prepared consisting of a two-lane road winding through mainly rural scenery, divided into four main sections of uninterrupted road, stretching out for 9.1 km on average (about 7 min of driving time) and separated by small villages. Roads in each section consisted mainly of easy curves (about 80%) with a constant radius of 380 m and ranging in length from 120 m to 800 m. The road surface was marked on the edges by a continuous line (20 cm) and in the center by a discontinuous line (15 cm), and outside the edges, a soft shoulder was present. The participants drove the simulated car (width = 1.60 m) through the entire route twice, bringing the driven distance up to approximately 80 km. On the experimental sections, outside the villages, speed was controlled by the simulator and set to 80 km/h to prevent potential compensatory speed reactions to the experimental manipulations.

Support Triggers

The vehicle's LP, defined as the distance between the center of the participant's car and the middle of the (right-hand) driving lane, was sampled at 10 Hz during the entire experiment. Sampled LP values were further processed into three support trigger variables: (a) the proportion of driving time in the near-edge zones, (b) the proportion of driving time in the over-edge zones, and (c) SDLP (see Figure 2).

The near-edge zones were defined as strips 12.5 cm wide toward each lane edge and are illustrated by the gray areas in Figure 2a. Driving in this zone with any part of the vehicle counted as driving near the lane's edges, except if the car also entered the over-edge zone (see Figure 2b). Each second, the support algorithm counted the number of LP samples values inside these near-edge zones for the preceding 30 s and divided it by the total number of samples during this period (300 samples). Driving in the near-edge zones for more than 7.5 s (25%) triggered the HUD. For the second trigger variable, driving in the over-edge zones, the threshold was set to 3 s (10%). Finally, all 300 LP values were used to calculate SDLP, which triggered the HUD when more than 22 cm. During the adaptive support conditions, exceeding any of these thresholds activated the HUD, and support was deactivated

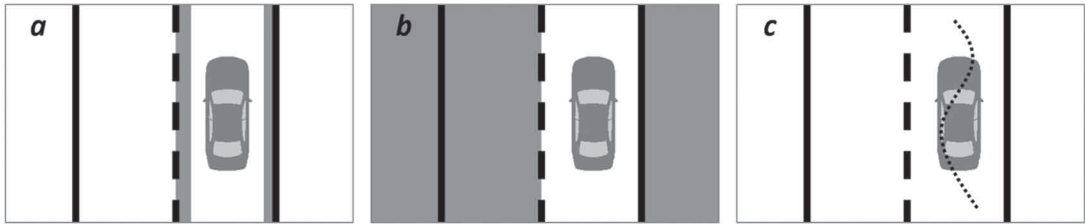


Figure 2. Abstract representation of the trigger variables. The gray areas in Figure 2a represent the near-edge zones toward the left and right lane edges, the gray areas in Figure 2b represent the over-edge zones on the left and right of the driving lane, and Figure 2c symbolizes the standard deviation of the lateral position.

when all trigger variables were below their threshold value. In addition, to prevent a high-frequency on-off switch, a minimum time delay of 10 s was set as a dead-band.

Driving near the lane's edges, driving over the lane's edges, and weaving within the lane were assumed to be an indication of reduced lateral control, increasing the likelihood of an accident and therefore a reason to provide lateral control support. However, for a working support system, the exact width of the near-edge zone, the time length of the dead-band, and the time window used to process LP data needed to be determined in addition to the threshold values. For SDLP, Brookhuis, de Waard, and Fairclough (2003) proposed an absolute deteriorated driving criterion of 25 cm. Such clear-cut indication of threshold values for the other trigger variables do not exist in literature. During a pilot study ($n = 8$), a number of calibrations were tested. However, given the large number of possible calibrations, the final values were based on the rather intuitive feeling of "rightness." The support system was therefore calibrated in a way that most participants reported that their lateral control had worsened in the period before the support system started supplying information on LP.

Design and Procedure

Each participant completed all conditions and was instructed to drive as he or she would normally do and to follow automated auditory instructions when driving. The order of the conditions was balanced according to the Latin square method, except for the low- and high-traffic-density conditions, which alternated.

Approximately 6 min of low-density oncoming traffic was always directly followed by a relatively short period of 65 s during which oncoming traffic was intensified. Hereafter, approximately 300 m before entering a village, the longitudinal control was returned to the driver. In each village, the participants received the instruction to pull over and park the car, followed by a 2-min break during which they were requested to complete mental effort rating scales.

The effects of three support modes were compared. On each road section, the HUD was either turned off (the no-support mode), continuously activated (the nonadaptive-support mode), or triggered when a threshold value was exceeded (the adaptive-support mode). In addition, lane width was either 2.25 m or 3.00 m for each road section. During the last 1,300 m of each road section, the average interval between oncoming vehicles decreased from 10 s to 1.5 s. No traffic was present on the same (driving) lane as the participants'. This resulted in a within-subject design consisting of three repeated measures factors: support mode (3), lane width (2), and oncoming traffic density (2).

Before the actual experiment started, participants were asked to read information related to the support system; this information included a full and detailed user's manual of the HUD and the exact trigger-related properties of the adaptive support mode. The main reason for providing this level of information beforehand was to make sure that the participants would understand the support system and avoid "automation surprise" (e.g., Prinzl, 2002). Furthermore, it was stressed that any HUD activation as a result of driving behavior did not indicate bad steering behavior but was

meant to provide LP information. After providing informed consent, the participants were given the opportunity to get used to the simulator car and all support modes. The Ethical Committee of the Psychology Department of the University of Groningen approved the study.

User Experience Questionnaire

After the participants were informed about the functionality of the HUD in both support modes, participants were asked to complete a technology acceptance questionnaire that consist of nine 5-point rating items that load on two factors, usefulness and satisfaction, to compare both support modes (Van der Laan, Heino, & de Waard, 1997). In addition, these expectation scores were compared with the scores given at the end of the experiment after the drivers had experienced both support types.

To detect possible differences in the support usage among the participants, they were asked to indicate in what way they had mainly used the HUD directly after the experiment. Three standard options were given: (a) "I used the lateral position information provided by the HUD," (b) "I used the HUD activation as a signal to improve lateral position behaviour," and (c) "I ignored the HUD as much as possible." In addition, the participants could tick a fourth option, "Different, namely . . .," and provide their own answer.

During each driving break, the participants were asked to mark the Rating Scale Mental Effort (RSME; Zijlstra, 1993) for both the first part (low-density oncoming traffic) and the last part of the drive (high-density oncoming traffic). The RSME is a single-dimension rating scale ranging from 0 to 150 that can be used by the participants to rate experienced effort. In addition, several effort indications (calibrated anchor points) are visible alongside the scale that may further guide the participant's rating. Indications start with *absolutely no effort* (RSME score of 2) and end with *extreme effort* (RSME score of 112).

Analyses

We analyzed the data using the General Linear Model Repeated Measures test of SPSS. Repeated-measures MANOVAs were run on

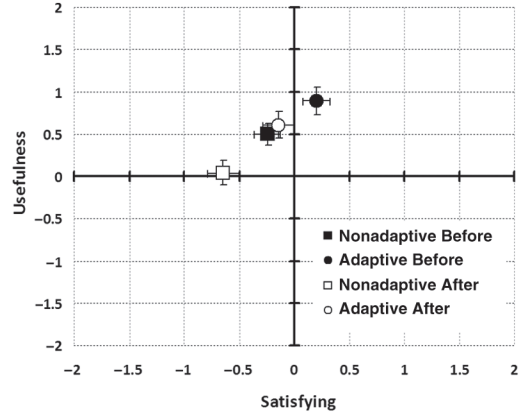


Figure 3. System acceptance scores before and after the experiment for both support modes. Error bars represent the standard error.

the system acceptance scores (usefulness and satisfaction) and the lateral control variables. We tested LP, SDLP, and time in the near- and over-edge zones using the mean value for all time windows within an experimental condition. ANOVAs were run on the RSME scores. Alpha was set to 5% for all tests, and to compare individual support modes, Bonferroni post hoc tests were used when appropriate.

RESULTS

User Experiences

To begin with, there were differences among participants in the way the HUD was mainly used. Of all the participants, 39% indicated that they ignored the HUD as much as possible. Furthermore, within the group of participants who did not ignore the HUD, 79% used the HUD primarily as a warning signal, whereas 16% reported to have used it as a source of LP information.

The subjective scores for both the usefulness and the satisfaction dimension (both ranging from -2 to +2) are shown in Figure 3. When looking at these two dimensions in Figure 3, one can see that the usefulness scores deviate most from the neutral, zero line and that adaptive support prior to the experiment received the highest score. On the satisfaction dimension, three out of four scores are within 0.5 points of the zero line. Despite these small ranges, both usefulness and

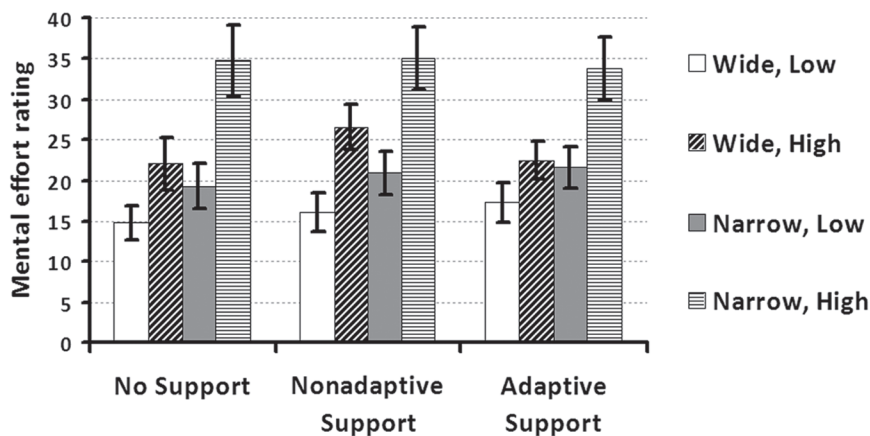


Figure 4. Reported mental effort for all support modes. Wide = wide lane (3.00 m); narrow = narrow lane (2.25 m); low = low-density oncoming traffic; high = high-density oncoming traffic. Error bars represent the standard error. Maximum score for mental effort is 150.

satisfaction show two main effects (support type, and before and after testing). Overall, adaptive support was rated higher than the nonadaptive support on both usefulness and satisfaction dimensions: usefulness, $F(1, 29) = 9.91, p < .01, \eta_p^2 = .26$; satisfaction, $F(1, 29) = 16.02, p < .001, \eta_p^2 = .36$. Also, the participants' expectation scores on both dimensions were higher than were the scores after experiencing either support type: usefulness, $F(1, 29) = 10.79, p < .01, \eta_p^2 = .27$; satisfaction, $F(1, 29) = 14.34, p < .01, \eta_p^2 = .33$.

The mental effort ratings are shown in Figure 4. Mean mental effort ranges from just over the *almost no effort* mark for wide lane, no support, and low-density traffic to just below the *some effort* mark for narrow lane, no support, and high-density traffic. Several effects on mental effort were present in the data. To begin with, the wide lane conditions were rated significantly lower than the narrow lane conditions (19.9 vs. 27.6 points). Also, low-density oncoming traffic was rated lower than the high-density conditions (18.3 vs. 29.1 points). However, the differences between the support modes did not result in a significant effect (10.5, 11.9, and 11.1 points for no support, nonadaptive support, and adaptive support, respectively). Finally, one interaction effect is present. When participants were driving on a wide lane, an increase of traffic density resulted in an increase of 7.6 RSME

TABLE 1: Univariate Test Results for Subjective Effort (Figure 4)

Effect	<i>F</i> (<i>df</i> 1, <i>df</i> 2)	<i>p</i>	η_p^2
Support (S)	<1 (2, 58)	<i>ns</i>	.005
Width (W)	11.58 (1, 29)	.002	.285
Traffic (T)	26.63 (1, 29)	<.001	.479
S $\times$ W	<1 (2, 58)	<i>ns</i>	.007
S $\times$ T	1.46 (2, 58)	<i>ns</i>	.048
W $\times$ T	11.24 (1, 29)	.002	.279
S $\times$ W $\times$ T	<1 (2, 58)	<i>ns</i>	.033

Note. Support = support mode; width = lane width; traffic = oncoming traffic density. Significant effects ($p < .05$) are shown in bold. Degrees of freedom are Greenhouse-Geiser corrected when Mauchly's test showed violation of sphericity.

points. When they were driving on a narrow lane, this increase was 14.0 points (see Figure 4 and Table 1).

Performance Measures

For LP, defined as the distance between the center of the participant's car and the middle of the (right-hand) driving lane, main effects for all factors were found (see Figure 5a and Table 2). Driving on a narrow lane and exposure to high-density oncoming traffic were associated with

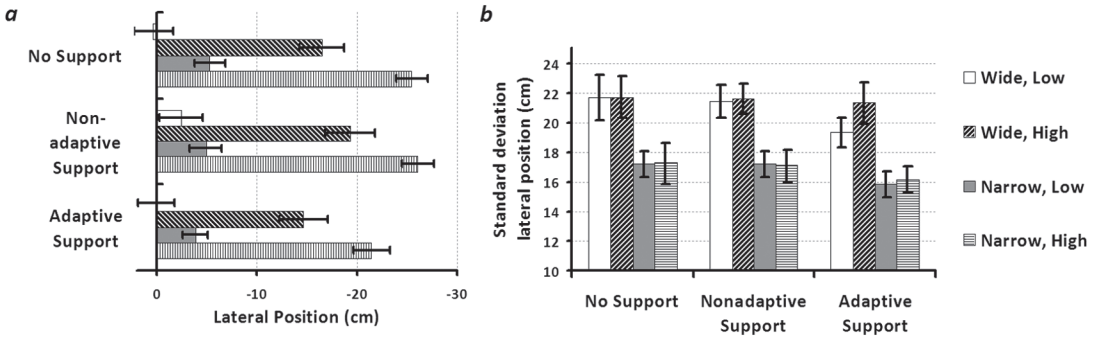


Figure 5. Mean (a) and standard deviation (b) of the lateral position for all support modes. Lateral position values represent the middle of the car (car width = 1.60 m) in relation to the middle of the right (driving) lane. Negative numbers indicate a position to the right of the lane middle and vice versa. Wide = wide lane (3.00 m); narrow = narrow lane (2.25 m); low = low-density oncoming traffic; high = high-density oncoming traffic. Error bars represent the standard error.

an LP toward the shoulder. In addition, these effects strengthened each other, creating a significant interaction. Also, support mode affected LP. Bonferroni post hoc comparisons did not reveal any significant differences, although the largest contributor to the main effect is the difference between the adaptive and the nonadaptive support mode (mean difference = 3 cm, $p = .051$).

As can be seen in Figure 5b, driving on narrow lanes decreased SDLP values compared with driving on wide lanes. Also, support mode yielded a main effect. A closer look revealed that SDLP was lower for the adaptive support mode than for the no-support mode (mean difference = 2 cm, $p = .027$). Increases in traffic density did not change SDLP.

The time participants drove with any part of the vehicle within 12.5 cm of either edge of the driving lane (but not over the edge) was not affected by support mode (see Figure 6a and Table 2). However, participants did spend more time inside the near-edge zone when on narrow lanes and during high oncoming traffic density.

The last vehicle parameter refers to the time participants drove with any part of the car over the driving lane edges (Figure 6b), and main effects were found for all factors. An increase in time spent over the driving lanes was associated with both driving on a narrow lane and high-density oncoming traffic. However, these factors seemed

to reinforce each other, resulting in an interaction. Last, a difference was found between support modes. Bonferroni post hoc analysis revealed a significant difference between adaptive support and nonadaptive support (mean difference = 3%, $p = .017$).

DISCUSSION

This study was designed to investigate the driving effects of nonadaptive and adaptive driver support in a wide range of driving demands and to assess drivers' experiences with using these systems. Differences between support modes were smaller both in number and in size when compared with the effects caused by manipulations of infrastructure and traffic. Effort ratings were unaffected by support mode, although most vehicle parameters did show a difference. On closer examination, it seems that these effects were mainly caused by the adaptive support. Compared with either the nonadaptive or the no-support mode, participants drove more centrally, swerved less, and drove less on the shoulder. In addition, none of the post hoc comparisons between the nonadaptive and the no-support mode were statistically significant.

Instead, there were distinct differences in the manner in which the HUD was primarily used. More than a third of the participants indicated to have ignored the HUD. This behavior might be

TABLE 2: Univariate Test Results for Vehicle Parameters (Figures 5a, 5b, 6a, 6b)

Effect	Mean LP		SDLP			Inside Near-Edge Zone			Inside Over-Edge Zone		
	F(df1, df2)	p	η^2_p	F(df1, df2)	p	η^2_p	F(df1, df2)	p	η^2_p	F(df1, df2)	p
Support (S)	3.81 (2, 56)	.028	.120	3.90 (2, 56)	.026	.122	<1 (2, 37.82)	ns	.033	5.10 (2, 56)	.009
Width (W)	31.93 (1, 28)	<.001	.533	59.59 (1,28)	<.001	.680	218.75 (1, 28)	<.001	.887	146.41 (1, 28)	<.001
Traffic (T)	183.67 (1, 28)	<.001	.868	<1 (1,28)	ns	.016	3.40 (1, 28)	<.001	.521	106.42 (1, 28)	<.001
S x W	<1 (2, 56)	ns	.021	<1 (2,56)	ns	.005	<1 (1.5, 43.0)	ns	.000	2.83 (1.6, 44.1)	.068
S x T	2.63 (2, 56)	.081	.086	1.02 (2,56)	ns	.035	1.16 (2, 56)	ns	.040	1.14 (2, 56)	ns
W x T	9.03 (1, 28)	.006	.244	<1 (1,28)	ns	.032	<1 (1, 28)	ns	.015	94.91 (1, 28)	<.001
S x W x T	<1 (2, 56)	ns	.012	<1 (2,56)	ns	.023	<1 (2, 56)	ns	.027	<1 (2, 6)	ns

Note. Support = support mode; width = lane width; traffic = oncoming traffic density; LP = lateral position; SDLP = standard deviation lateral position. Significant effects ($p < .05$) are shown in bold. Degrees of freedom are Greenhouse-Geiser corrected when Mauchly's test showed violation of sphericity.

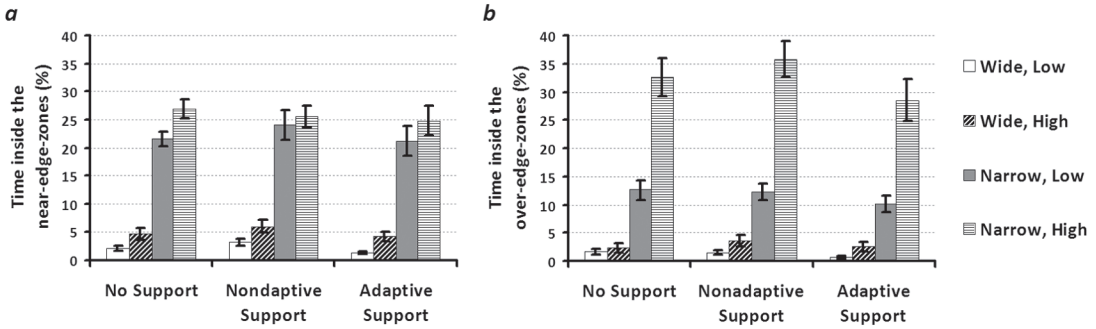


Figure 6. Percentage of the time spent inside the near-edge zone for wide lane (a) and over-edge zone (b) for all support modes. Wide = wide lane (3.00 m); narrow = narrow lane (2.25 m); low = low-density oncoming traffic; high = high-density oncoming traffic. Error bars represent the standard error.

classified as *disusing* the provided assistance, which is defined as a failure to engage in automation when it could improve performance (Lee, 2008; Parasuraman & Riley, 1997). What has caused these differences can only be speculated. In addition to design specifics, use of the system may correlate with personality traits relevant to driving behavior, such as sensation seeking (e.g., Constantinou, Panayiotou, Konstantinou, Loutsiou-Ladd, & Kapardis, 2011). Also, participants were not asked about their driving history in terms of traffic violations, accident involvement, and so on. For future research, it would be interesting to use this kind of information to establish who is more likely to ignore useful information. In addition, we cannot precisely discern which HUD version the participants had in mind when answering this question because it was asked only once, shortly after the experiment. However, the satisfaction and the usefulness scores indicate that the adaptive HUD was the most accepted.

Within the group of participants who did not mainly ignore the HUD, 79% used HUD activation as a warning signal; we can therefore conclude that the participants mainly appreciated the warning property of the adaptive HUD and not its information provision property. Although the participants did not use adaptive support as intended by the system's designers, it seemed to help driving performance. It could be argued that using naive participants might have prevented usage of the system in this way. However, this scenario is unlikely. If this system had been

tested across a longer period, naive drivers eventually would have learned when support is activated.

When comparing the current system with previously investigated adaptive systems, one can find several differences and similarities. Adaptive automation is usually applied to (modified versions of) complex tasks, such as the Multi-Attribute Task (e.g., Pope, Bogart, & Bartolome, 1995), the Cabin Air Management task (e.g., Ting et al., 2010), the Multitask (e.g., Kaber & Endsley, 2004), or a simulated helicopter cockpit task (Miller & Hannen, 1999). Adequate performance of these tasks requires a lot of participant training or trained professionals. In contrast, a driving simulator used for the current study closely resembles a task environment highly familiar to most people. Other comparisons can be made with regard to the aid itself. A focus of the current study was aiding the input functions of the human information processing system (Parasuraman et al., 2000), which it has in common with the Cockpit Information Manager (CIM; Miller & Hannen, 1999), even though the LOA in the current study must be considered lower since it did not involve organizing the data in any way.

In other studies, the output functions (decision making and response selection) are supported. For example, Ting et al. (2010) developed an adaptive aid that could control the regulation of crucial air parameters. Also, compensatory tracking, as used by Pope et al. (1995), was either fully automated or fully manual. Apropos,

given the focus on input functions and a relatively low LOA, the current system could be referred to as *assistance* rather than *automation* on the continuum of automation degrees presented in Flemisch et al. (2008).

Another relevant feature of the adaptive system described in the current article is the use of task performance to trigger aid, even though many systems were designed with a different approach in mind (a classification is given in Parasuraman et al., 1992). The CIM, for example, uses context information and crew actions to determine current and near-future tasks to adjust information provision. Another alternative approach was implemented by Ting et al. (2010), who combined electrocardiogram and electroencephalogram data to predict error performance, which was subsequently used to activate task support. The aforementioned comparisons represent just a selection of dimensions along which comparisons can be made. Also, more systems could have been mentioned. Regretfully, a more thorough classification of all published adaptive automated systems is beyond the scope of the current article.

A HUD was chosen as the information carrier because it increases gaze time toward the roadway environment compared with head-down displays. However, several problems with using a HUD have been mentioned in the literature. For example, the HUD's focal distance can influence perception time of both the information on the HUD and the roadway environment (Gish & Staplin, 1995; Tufano, 1997). Also, cluttering of the visual field, superimposing of salient environmental information, and poor detection of unexpected peripheral events as a result of attentional capture can be problematic (e.g., Horrey & Wickens, 2004; Weintraub & Ensing, 1992).

These problems were probably less relevant for the present study. To start with, driving in a rural environment and the HUD position (see Figure 1) minimized the superimposing of oncoming traffic. Also, the visual workload demanded by the driving situation can be considered quite low, making it unlikely that adding the HUD overloaded visual attentional resources. Last, peripheral detection was not an issue in the simulator, since salient peripheral objects were

not present. However, the issues associated with using HUDs are extremely relevant in a real-world application and might be overcome by using what has been called "conformal imagery," a form of augmented reality (Fadden, Ververs, & Wickens, 2001).

A limitation of the study may be the calibration of the adaptive support system. For the adaptive system, several parameters needed to be set, such as the threshold values, and extensive testing of the complete range of calibrations would have been a daunting enterprise. To solve this problem, we specifically aimed at avoiding a calibration that would make the adaptive system appear unfair or unjust as evaluated by the drivers during a pilot study. System calibration was partly based on trial and error but ultimately on the feeling of rightfulness by the pilot test participants. Using a different calibration for the adaptive support mode may have produced different experimental results. Speculatively, using stricter LP-related thresholds (driving time in the near- and over-edge zones) could have resulted in more HUD activations and therefore in driving more centrally. However, it is not clear how (de)activation frequency would interact with the willingness to use the provided information; this question may be a topic for future research.

Another limitation is the implementation of the high-density oncoming traffic conditions, which were shorter than the low-density conditions and occurred in a fixed, alternating order. Alternating the traffic density levels was done to prevent interference from the high-density condition in the quiet condition. It is unclear whether a different design would have produced different results. However, in hindsight, a balanced order and equal duration of these conditions would have been preferred to decrease the likelihood of order effects and to exclude the possibility that condition length confounded the large effects of increasing oncoming traffic density.

Last, only lane-keeping performance was used to assess MWL changes even though a change in performance does not always equate to a change in MWL. Changes in performance levels may also be the result of changed task demands, whereby performance might degrade even if the MWL level is sustained. Also, a depletion of energetic

and cognitive resources, despite intentions of good performance, might prevent further increase in MWL (e.g., Hockey, 1997). Considering the flexibility displayed in human behavior, one may conclude that it is unlikely that a reliable single measure for MWL exists and that several sources (subjective, performance, environment, and psychophysiology) are required for deriving a more reliable index (Brookhuis & de Waard, 2000; de Waard, 1996; Gaillard & Kramer, 2000; Hoozeboom & Mulder, 2004).

Using several sources, such as physiological triggers, for example, would have potentially increased the range of MWL detection to beyond the point at which performance cannot be increased anymore. However, as the intention was to assess changes in MWL, rather than absolute levels, it is carefully assumed that within a highly controlled experimental setting, a performance change will likely be caused by a change in resource allocation in service of the steering task and, therefore, a change in MWL.

CONCLUSION

Across a range of driving demands, driving performance was somewhat improved when drivers were exposed to an adaptive support system that provided feedback on LP. This was likely caused by unexpected use of the system. Participants indicated to have mainly used the moment of support activation as a warning signal rather than using the information provided on the HUD itself. In this way, the system not only supported the perception phase but supported the decision-making phase as well. It seems unlikely that an adaptive support system can be designed that will work only for the pre-decision stages; indicating that all perceived properties of an adaptive system are likely to be utilized by the human operator.

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KEY POINTS

- An adaptive feedback loop providing lateral position was preferred to a nonadaptive feedback loop mainly because drivers used the activation of the display as a warning sign.
- A third of the participants indicated to have ignored the head-up display carrying lateral position information.
- Only adaptive support resulted in an improvement of lateral control.
- Effects of a support initiation can be expected, regardless of the support type that is activated.

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